

Exam Distribution

Ad HOC Algorithms Engineering Contest 2026

Student life in movies looks like the dream: endless parties, deep conversations, and carefree nights. In reality, it is mostly caffeine, deadlines, and being sleep deprived. And just when you think you have hit rock bottom, the exam session arrives.

But while exams are challenging for students, we should not forget about the problems that university staff face during the exam period! They must assign students to different seats so that (1) students who take the same exam should be in as few rooms as possible, (2) students in the same room who take the same exam should sit as close together as possible, and (3) notorious cheaters should not be sitting near other students who take the same exam. As the number of students is rising significantly and the university staff is heavily overworked, they could really use your help with this!

Problem Description

During the examination period, the university must assign a large number of students to seats in a limited set of available rooms. Each room contains a fixed rectangular seating layout, and students belong to different exam groups, where we simply assume that students make the same exam if and only if they are in the same exam group. Ideally, students taking the same exam should be placed together in as few rooms as possible to reduce logistical overhead.

Additionally, some students are marked as *cheaters*. To reduce the opportunities for cheating, a cheater should not be seated next to another student taking the same exam. Concretely, if a listed cheater has a student from the same exam group in one of the eight neighboring seats, horizontally, vertically, or diagonally, a fixed penalty is incurred for each such occurrence.

Finally, invigilators prefer not to walk long distances inside rooms. As each invigilator can handle only one exam group within one room, we also want to minimize the longest distance between two students taking the same exam within the same room.

Input

The input has the following structure:

- The first line contains three integers N ($N \leq 10\,000$), M ($M \leq 100$), and G ($G \leq 100$), denoting the number of students, rooms, and exam groups, respectively.
- The second line contains three non-negative coefficients:

$$c_{\text{cheat}}, c_{\text{split}}, c_{\text{diam}}.$$

These coefficients control the relative importance of cheating prevention, group splitting, and compact seating arrangements.

- The next N lines describe the (0-based) exam group of each student i :

$$g_i \in \{0, \dots, G-1\}.$$

The order of these lines matches the student indices. That is, the first such line describes student 0, the second describes student 1, and so on.

- The following M lines describe the rooms. For room j , the line contains:

$$row_j, col_j,$$

where row_j and col_j are the number of rows and columns in the room.

- The next line contains one integer C , the number of listed cheaters.
- Finally, C lines follow, each containing one integer:

$$u.$$

Here u is the 0-based index of a student marked as a cheater.

Output

The output should describe the seating assignment and must contain exactly N lines. Line $i + 1$ describes the seat assigned to student i and contains three integers:

$$m_i, r_i, c_i.$$

Here m_i is the (0-based) index of the room assigned to student i , and r_i and c_i are the (0-based) row and column of the assigned seat inside that room.

A solution is valid if and only if:

- each student is assigned to exactly one seat,
- every assigned seat is inside the bounds of its room,
- no two students are assigned to the same seat.

Cost function

The cost function consists of two components: the invigilation cost f_I and the cheating cost f_C .

Invigilation cost. Each exam group g needs a separate invigilator for each room m in which at least one student is taking the corresponding exam. For each invigilator we pay a fixed cost plus some cost depending on the maximum distance between two students of group g in room m .

Specifically, let M_g be the set of rooms in which there is at least one student of group g . For some room $m \in M_g$, let $S_{g,m}$ be the set of seats in room m taken by students of group g :

$$S_{g,m} = \{(r_i, c_i) \mid \text{student } i \text{ is in group } g \text{ and seated in room } m\}.$$

We can compute the diameter of this set simply as follows:

$$\text{diam}(S_{g,m}) = \max_{(r_i, c_i), (r_j, c_j) \in S_{g,m}} \|(r_i, c_i) - (r_j, c_j)\|_2,$$

where $\|\cdot\|_2$ denotes the Euclidean norm. The invigilation cost for group g in room m is defined as:

$$f_I(g, m) = c_{\text{split}} + c_{\text{diam}} \text{diam}(S_{g,m}).$$

The total invigilation cost is the sum over all invigilators divided by the number of exam groups:

$$f_I = \frac{1}{G} \sum_{g=0}^{G-1} \sum_{m \in M_g} f_I(g, m)$$

Cheating cost. For each cheater we assign a penalty for each student making the same exam in one of the eight neighboring seats. Note that we assign the penalty twice if two cheaters of the same exam group are sitting in neighboring seats (once for each cheater).

Specifically, let Q be the set of student indices of cheaters. For a cheater $q \in Q$, let Z_q be the set of student indices that are too close to q and making the same exam, that is:

$$Z_q = \{i \neq q \mid g_i = g_q \wedge m_i = m_q \wedge |r_i - r_q| \leq 1 \wedge |c_i - c_q| \leq 1\}.$$

The total cheating cost is then defined as follows:

$$f_C = \frac{c_{\text{cheat}}}{C} \sum_{q \in Q} |Z_q|,$$

where $f_C = 0$ if the number of cheaters $C = 0$.

Total cost. The total cost of a solution is simply the sum of the invigilation and cheating cost:

$$F = f_I + f_C$$

This cost should be **minimized**.